LibreOffice 26.2 Help

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Statistical Functions Part Three

CONFIDENCE

Returns the (1-alpha) confidence interval for a normal distribution.

Syntax

```
CONFIDENCE(Alpha; StDev; Size)
```

Alpha is the level of the confidence interval.

StDev is the standard deviation for the total population.

Size is the size of the total population.

Example

`=CONFIDENCE(0.05;1.5;100)` gives 0.29.

CONFIDENCE.NORM

Returns the (1-alpha) confidence interval for a normal distribution.

Syntax

```
CONFIDENCE.NORM(Alpha; StDev; Size)
```

Alpha is the level of the confidence interval.

StDev is the standard deviation for the total population.

Size is the size of the total population.

Example

`=CONFIDENCE.NORM(0.05;1.5;100)` gives 0.2939945977.

Technical information

This function is available since LibreOffice 4.2.

This function is **NOT** part of the **Open Document Format for Office Applications (OpenDocument) Version 1.3. Part 4: Recalculated Formula (OpenFormula) Format** standard. The name space is

`COM.MICROSOFT.CONFIDENCE.NORM`

CONFIDENCE.T

Returns the (1-alpha) confidence interval for a Student's t distribution.

Syntax

`CONFIDENCE.T(Alpha; StDev; Size)`

Alpha is the level of the confidence interval.

StDev is the standard deviation for the total population.

Size is the size of the total population.

Example

`=CONFIDENCE.T(0.05;1.5;100)` gives 0.2976325427.

Technical information

This function is available since LibreOffice 4.2.

This function is **NOT** part of the **Open Document Format for Office Applications (OpenDocument) Version 1.3. Part 4: Recalculated Formula (OpenFormula) Format** standard. The name space is

`COM.MICROSOFT.CONFIDENCE.T`

CORREL

Returns the correlation coefficient between two data sets.

Syntax

`CORREL(Data1; Data2)`

Data1 is the first data set.

Data2 is the second data set.

Example

`=CORREL(A1:A50;B1:B50)` calculates the correlation coefficient as a measure of the linear correlation of the two data sets.

COVAR

Returns the covariance of the product of paired deviations.

Syntax

```
COVAR(Data1; Data2)
```

Data1 is the first data set.

Data2 is the second data set.

Example

`=COVAR(A1:A30;B1:B30)`

COVARIANCE.P

Returns the covariance of the product of paired deviations, for the entire population.

Syntax

```
COVARIANCE.P(Data1; Data2)
```

Data1 is the first data set.

Data2 is the second data set.

Example

`=COVARIANCE.P(A1:A30;B1:B30)`

Technical information

> This function is available since LibreOffice 4.2.

This function is **NOT** part of the **Open Document Format for Office Applications (OpenDocument) Version 1.3. Part 4: Recalculated Formula (OpenFormula)**

Format standard. The name space is

```
COM.MICROSOFT.COVARIANCE.P
```

COVARIANCE.S

Returns the covariance of the product of paired deviations, for a sample of the population.

Syntax

```
COVARIANCE.S(Data1; Data2)
```

Data1 is the first data set.

Data2 is the second data set.

Example

```
=COVARIANCE.S(A1:A30;B1:B30)
```

Technical information

This function is available since LibreOffice 4.2.

This function is **NOT** part of the **Open Document Format for Office Applications (OpenDocument) Version 1.3. Part 4: Recalculated Formula (OpenFormula) Format** standard. The name space is

```
COM.MICROSOFT.COVARIANCE.S
```

CRITBINOM

Returns the smallest value for which the cumulative binomial distribution is greater than or equal to a criterion value.

Syntax

```
CRITBINOM(Trials; SP; Alpha)
```

Trials is the total number of trials.

SP is the probability of success for one trial.

Alpha is the threshold probability to be reached or exceeded.

Example

`=CRITBINOM(100;0.5;0.1)` yields 44.

KURT

Returns the kurtosis of a data set (at least 4 values required).

Syntax

```
KURT(Argument 1 [; Argument 2 [; … [; Argument 255]]])
```

Argument 1; Argument 2; … ; Argument 255 are numbers, text, references to cells or to cell ranges of numbers or text.

The parameters should specify at least four values.

> This function ignores any text or empty cell within a data range. If you suspect wrong results from this function, look for text in the data ranges. To highlight text contents in a data range, use the value highlighting feature.

Example

`=KURT(A1;A2;A3;A4;A5;A6)`

LARGE

Returns the Rank_c-th largest value in a data set.

> This function is part of the Open Document Format for Office Applications (OpenDocument) standard Version 1.2. (ISO/IEC 26300:2-2015)

Syntax

```
LARGE(Data; RankC)
```

Data is the cell range of data.

RankC is the ranking of the value. If RankC is an array, the function becomes an array function.

> This function ignores any text or empty cell within a data range. If you suspect wrong results from this function, look for text in the data ranges. To highlight text contents in a data range, use the value highlighting feature.

Example

`=LARGE(A1:C50;2)` gives the second largest value in A1:C50.

`=LARGE(A1:C50;B1:B5)` entered as an array function gives an array of the c-th largest value in A1:C50 with ranks defined in B1:B5.

LOGINV

Returns the inverse of the lognormal distribution.

Syntax

```
LOGINV(Number [; Mean [; StDev]])
```

Number (required) is the probability value for which the inverse standard logarithmic distribution is to be calculated.

Mean (optional) is the arithmetic mean of the standard logarithmic distribution (defaults to 0 if omitted).

StDev (optional) is the standard deviation of the standard logarithmic distribution (defaults to 1 if omitted).

Example

`=LOGINV(0.05;0;1)` returns 0.1930408167.

LOGNORM.DIST

Returns the values of a lognormal distribution.

Syntax

```
LOGNORM.DIST(Number; Mean; StDev; Cumulative)
```

Number (required) is the probability value for which the standard logarithmic distribution is to be calculated.

Mean (required) is the mean value of the standard logarithmic distribution.

StDev (required) is the standard deviation of the standard logarithmic distribution.

Cumulative (required) = 0 calculates the density function, Cumulative = 1 calculates the distribution.

Example

`=LOGNORM.DIST(0.1;0;1;1)` returns 0.0106510993.

Technical information

This function is available since LibreOffice 4.3.

This function is **NOT** part of the **Open Document Format for Office Applications (OpenDocument) Version 1.3. Part 4: Recalculated Formula (OpenFormula) Format** standard. The name space is

```
COM.MICROSOFT.LOGNORM.DIST
```

LOGNORMDIST

Returns the values of a lognormal distribution.

Syntax

```
LOGNORMDIST(Number [; Mean [; StDev [; Cumulative]]])
```

Number is the probability value for which the standard logarithmic distribution is to be calculated.

Mean (optional) is the mean value of the standard logarithmic distribution.

StDev (optional) is the standard deviation of the standard logarithmic distribution.

Cumulative (optional) = 0 calculates the density function, Cumulative = 1 calculates the distribution.

Example

`=LOGNORMDIST(0.1;0;1)` returns 0.01.

LOGNORM.INV

Returns the inverse of the lognormal distribution.

This function is identical to LOGINV and was introduced for interoperability with other office suites.

Syntax

```
LOGNORM.INV(Number ; Mean ; StDev)
```

Number (required) is the probability value for which the inverse standard logarithmic distribution is to be calculated.

Mean (required) is the arithmetic mean of the standard logarithmic distribution.

StDev (required) is the standard deviation of the standard logarithmic distribution.

Example

`=LOGNORM.INV(0.05;0;1)` returns 0.1930408167.

Technical information

This function is available since LibreOffice 4.3.

This function is **NOT** part of the **Open Document Format for Office Applications (OpenDocument) Version 1.3. Part 4: Recalculated Formula (OpenFormula) Format** standard. The name space is

```
COM.MICROSOFT.LOGNORM.INV
```

SMALL

Returns the Rank_c-th smallest value in a data set.

This function is part of the Open Document Format for Office Applications (OpenDocument) standard Version 1.2. (ISO/IEC 26300:2-2015)

Syntax

```
SMALL(Data; RankC)
```

Data is the cell range of data.

RankC is the rank of the value. If RankC is an array, the function becomes an array function.

This function ignores any text or empty cell within a data range. If you suspect wrong results from this function, look for text in the data ranges. To highlight text contents in a data range, use the value highlighting feature.

Example

`=SMALL(A1:C50;2)` gives the second smallest value in A1:C50.

=SMALL(A1:C50;B1:B5) entered as an array function gives an array of the c-th smallest value in A1:C50 with ranks defined in B1:B5.

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